Structured Analysis and Structured Design (SA/SD)

Software Component Cataloguing System

Prepared by: Priyank Bansal

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Contents

1 Description

This document presents the Structured Analysis and Structured Design (SA/SD) of the Software Component Cataloguing System. It focuses on the system structure, flow of information, module boundaries, and maintainable design.

1.1 Purpose

The purpose of this document is to describe the software in a structured form so that the relationship between functions, processes, data, and modules can be clearly understood.

1.2 Scope

The system is a web-based catalogue for reusable software components. It supports registration, login, component storage, catalogue organization, searching, and usage tracking.

1.3 Maintenance and Context

The design must remain easy to update when new component fields, search filters, catalogue categories, or reporting features are added. The application should therefore be divided into clear modules with minimal dependency between them.

2 Development Plan

2.1 Analysis Phase

The system is first analyzed by identifying the external user, the major functions, and the data exchanged between the user and the system. The main outcome of this phase is the functional breakdown and the context-level view.

2.2 Design Phase

The analyzed functions are then converted into modules, data structures, and deployment layers. The main goal is to keep the design simple, modular, and easy to maintain.

2.3 Maintenance Plan

The design should support future enhancement without major restructuring. For example, additional metadata fields or improved search logic should be added with small changes only.

3 Target Users

- **User / Developer**: registers, logs in, searches components, and manages catalogue entries.
- **Admin**: oversees stored records and monitors catalogue-related information.

4 Structured Analysis and Structured Design (SA/SD)

4.1 Structured Analysis: Data Flow Diagrams (DFD)

The Structured Analysis utilizes Data Flow Diagrams to represent functional decomposition.

4.1.1 Level 0: Context Diagram

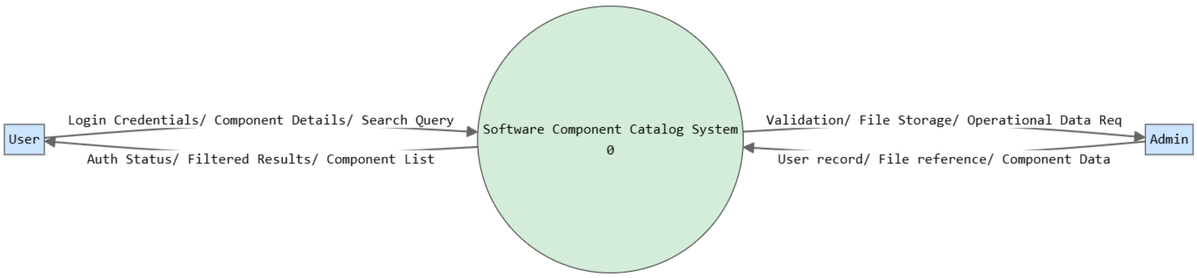


Figure 1: System Context Diagram (Level 0)

4.1.2 Level 1: Data Flow Diagram

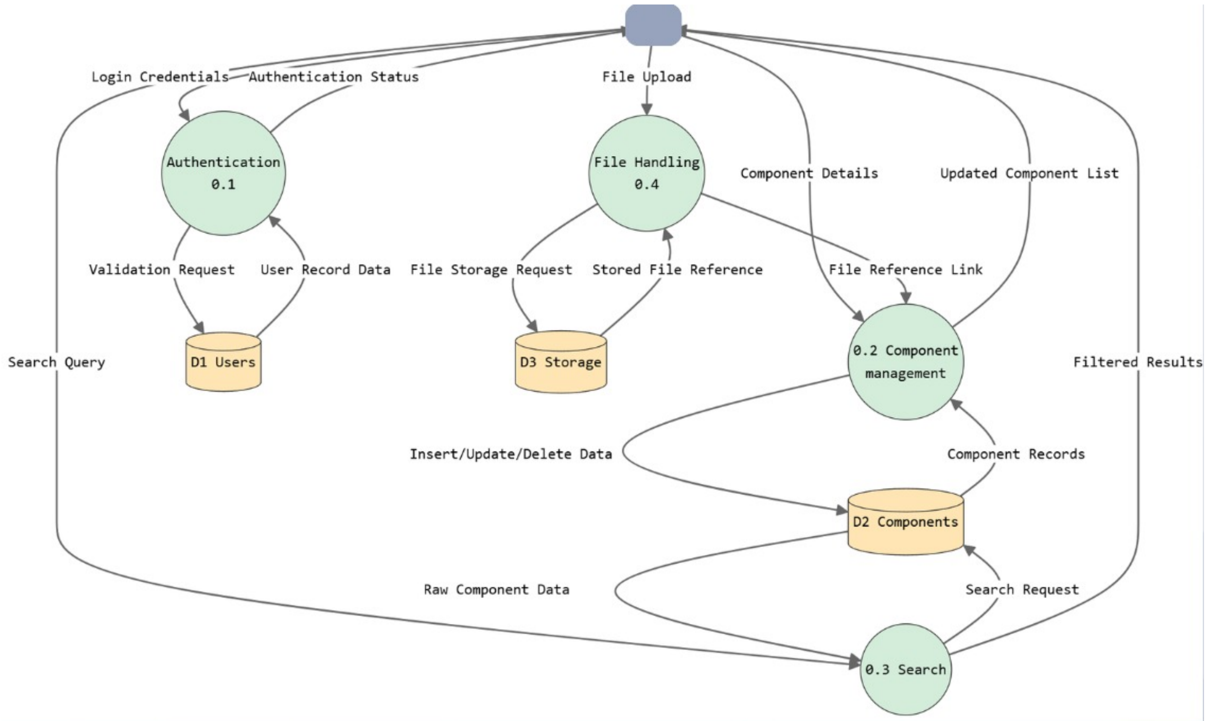


Figure 2: Data Flow Diagram (Level 1)

4.2 Level 2: Structured Design

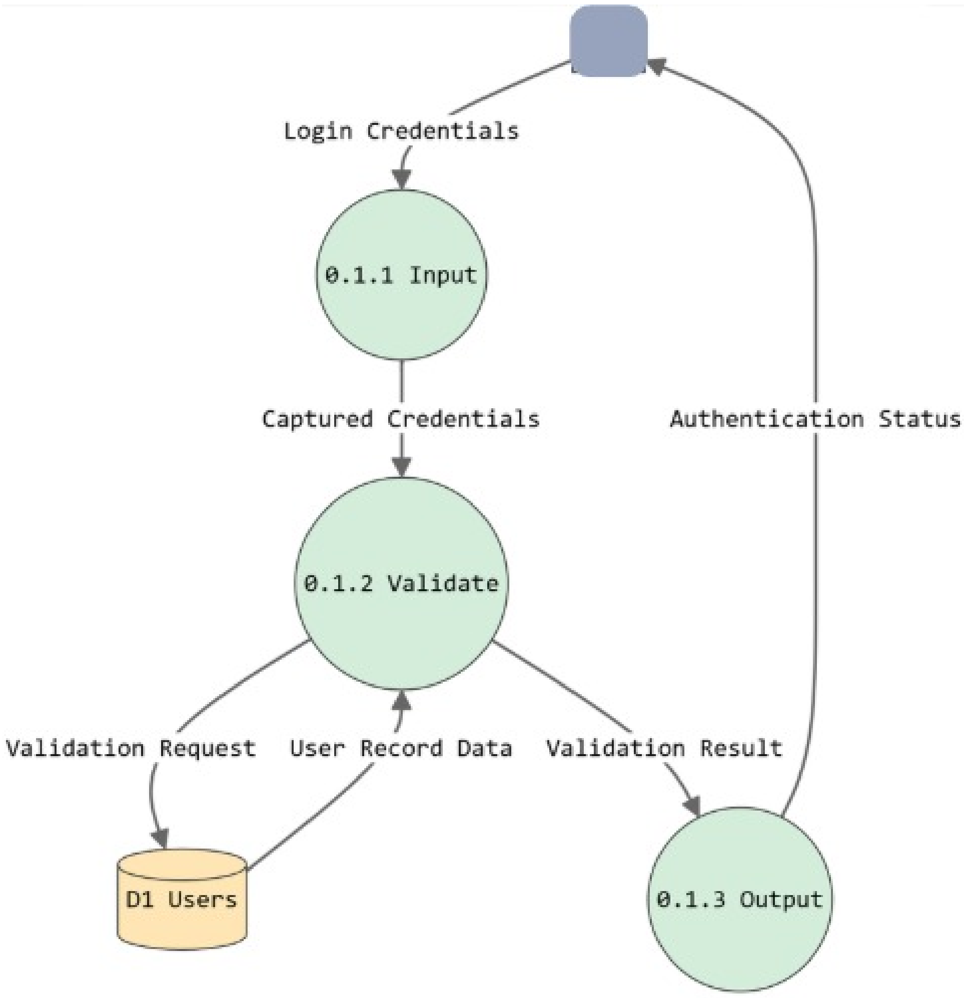


Figure 3: Level 2 - Authentication Module

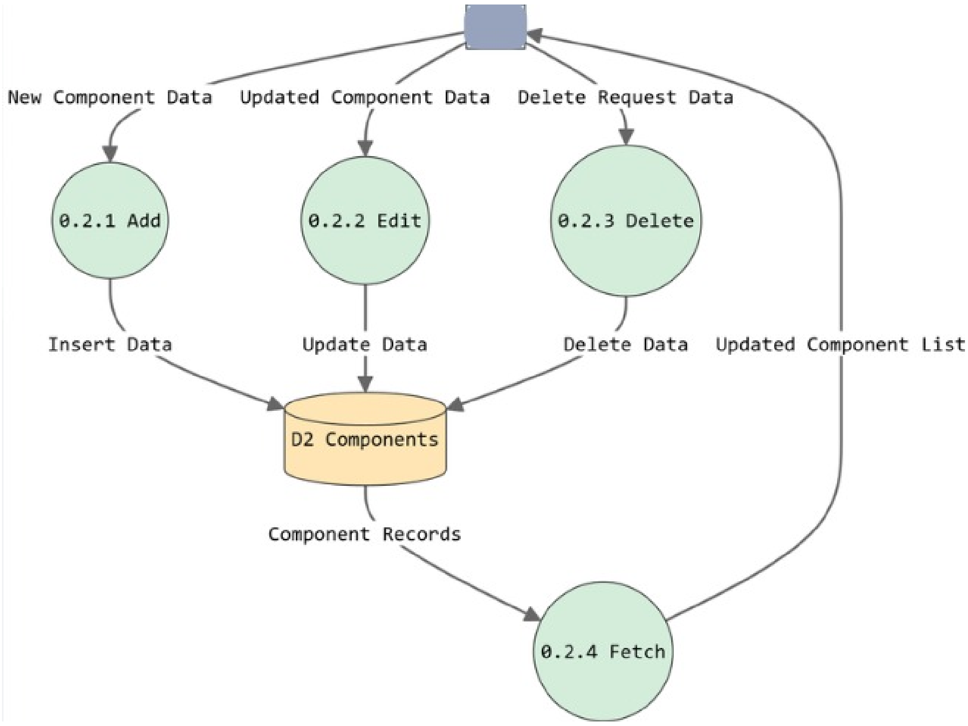


Figure 4: Level 2 - Component Management Module

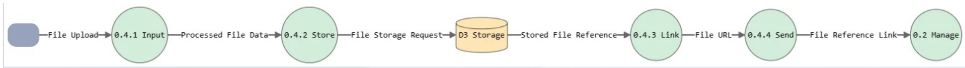


Figure 5: Level 2 - File Handling Module

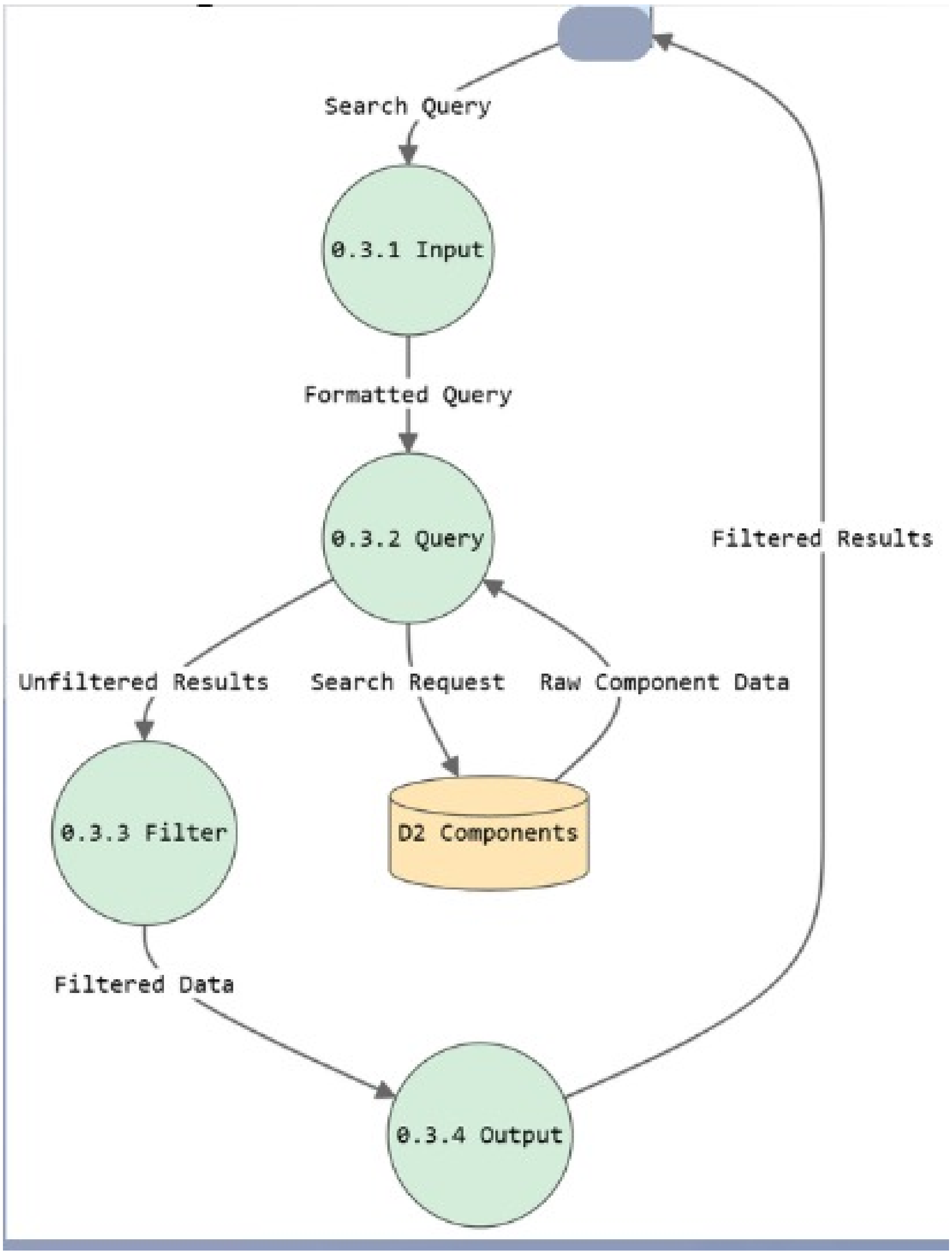


Figure 6: Level 2 - Search Module

4.3 Process Flow Descriptions

Authentication Flow

1. The user enters login or registration information.
2. The system validates the input.
3. If valid, the user is authenticated.
4. If invalid, an error message is displayed.

Add Component Flow

1. The user opens the component entry page.
2. The user enters component details.
3. The system validates the data.
4. Valid data is stored in the database.
5. A confirmation message is displayed.

Search Component Flow

1. The user enters a search query.
2. The system searches the stored records.
3. Matching records are retrieved.
4. The results are displayed to the user.

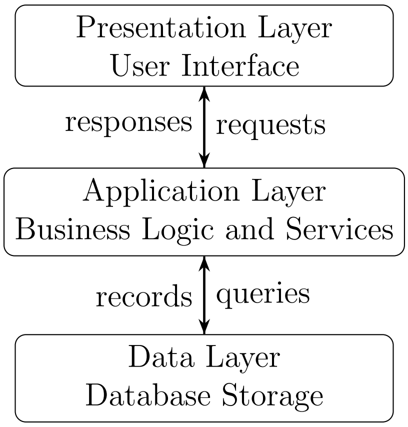
4.4 Data Dictionary

Data Item	Type	Description
UserID	Integer	Unique identifier for each user
Username	String	Login and display name
Email	String	User email address
Password	String	Authentication credential
CatalogueID	Integer	Unique identifier for a catalogue
CatalogueName	String	Name of the catalogue
ComponentID	Integer	Unique identifier for a component
ComponentName	String	Name of the software component
Description	Text	Component details and notes
Keywords	String	Search terms for retrieval
UsageCount	Integer	Number of times a component is used

5 Technologies

5.1 System Architecture

The system follows a three-tier architecture.



5.2 Module Design

The system is organized into the following modules:

- **Authentication Module**: handles registration and login.
- **Catalogue Module**: manages catalogue creation and organization.
- **Component Module**: stores and updates component records.
- **Search Module**: handles keyword search and result retrieval.
- **Usage Module**: tracks component access and reuse.

5.3 Control Flow

Login Control Flow

1. Enter credentials.
2. Validate credentials.
3. Grant or deny access.

Component Control Flow

1. Enter component details.
2. Validate details.
3. Save or reject the record.

Search Control Flow

1. Accept search query.
2. Match query against stored data.
3. Return matching results.

6 Distribution Plan

6.1 Deployment View

The system is deployed as a web application.

- Users access the system through a browser.
- The frontend communicates with backend services.
- The backend stores data in the database.

6.2 Maintenance Plan

- The modular structure supports future extension.
- New component attributes can be added with minimal changes.
- Search and reporting logic can be improved independently.

7 UML Diagrams

7.1 Use case diagram

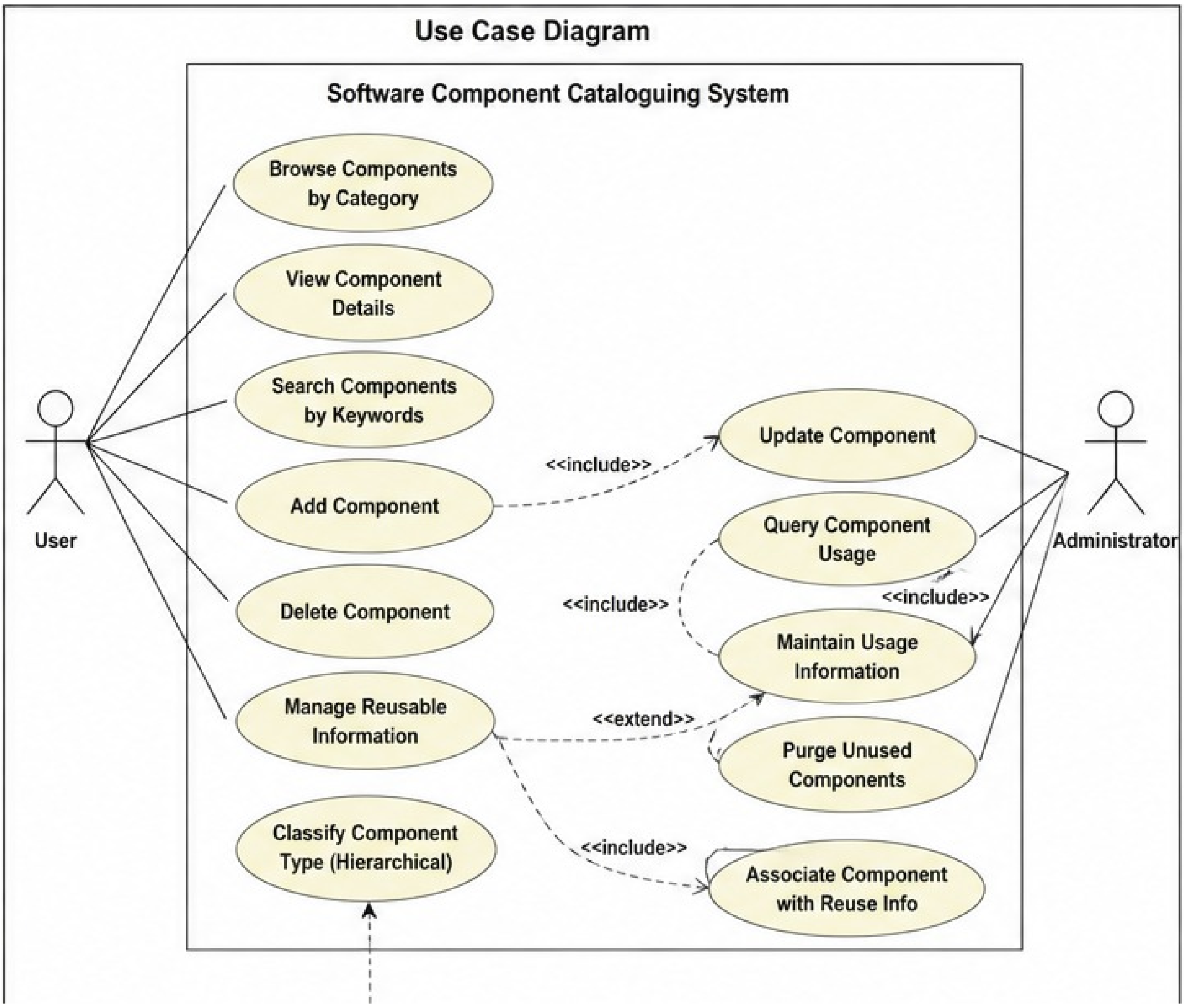


Figure 7: Use case diagram

7.2 Class diagram

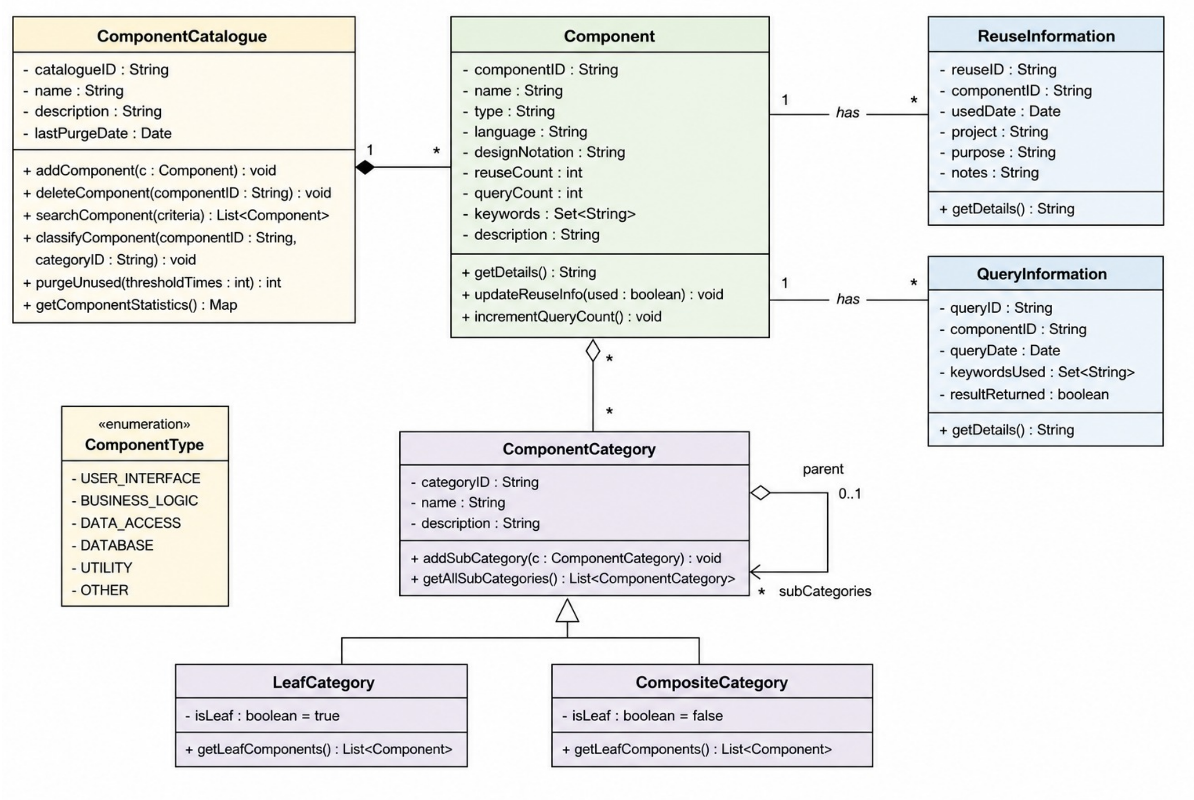


Figure 8: Class diagram

7.3 Sequence diagram

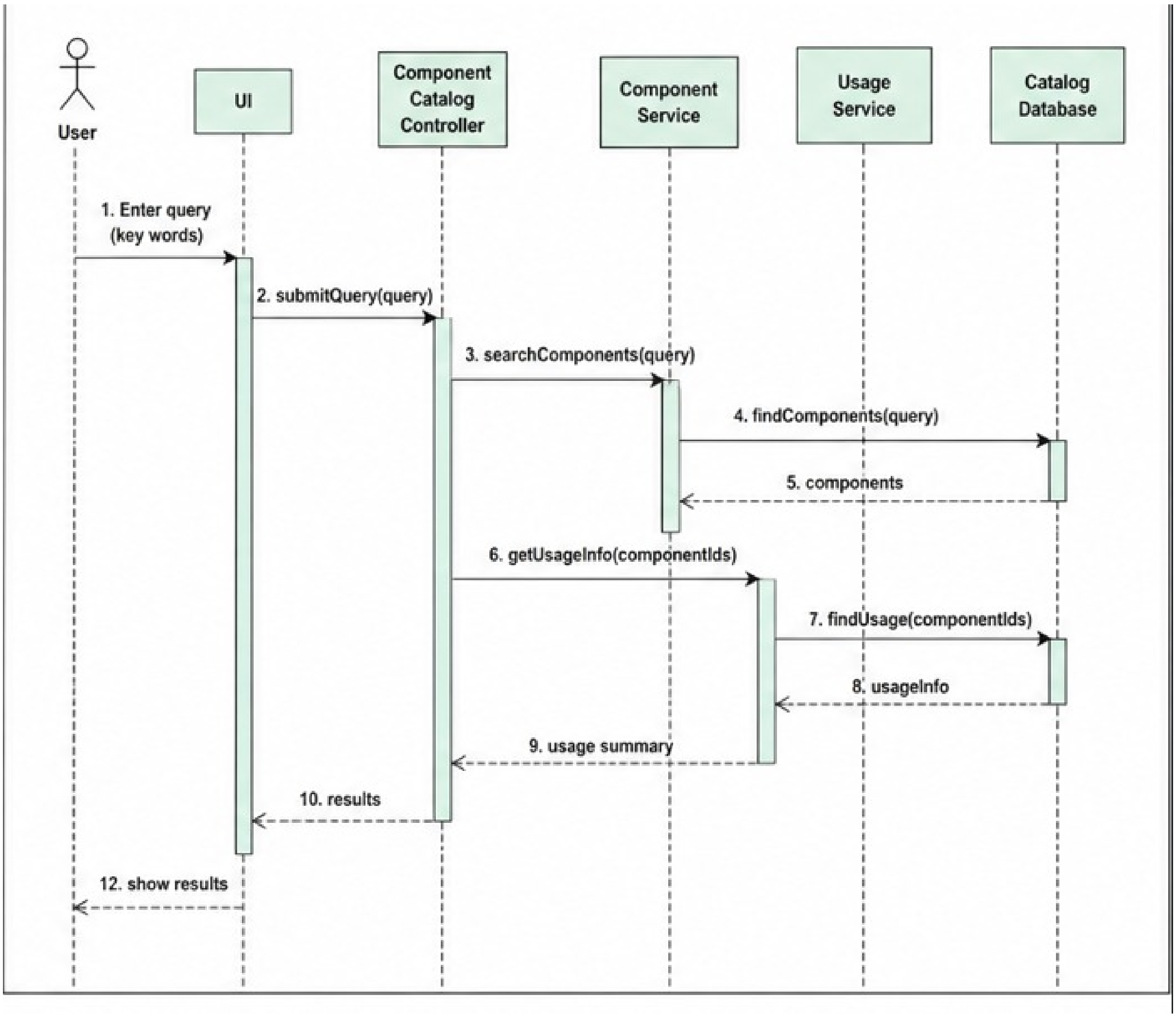


Figure 9: Sequence diagram

7.4 State chart diagram

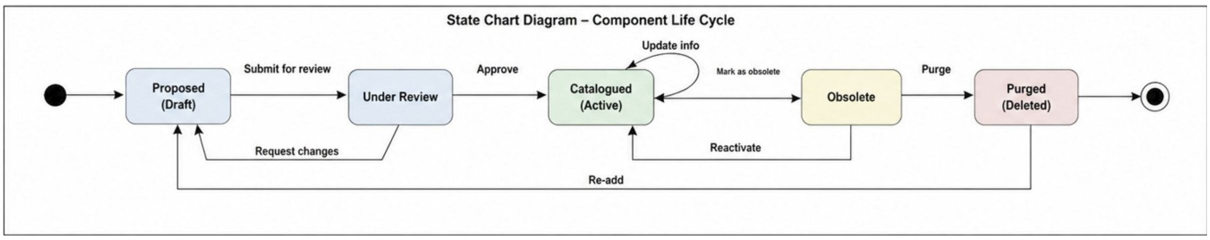


Figure 10: State chart diagram

8 Conclusion

This SA/SD document provides the structural view of the Software Component Cataloguing System. It includes the functional decomposition, context diagram, DFD Level 1, process flows, data dictionary, architecture, module design, and maintenance planning needed for submission.